

Let G be a topological group, let $\mathfrak{L}(G)$ be the space of its closed subgroups equipped with the E -topology. The open prebase of this topology is formed by the sets $D_1(U) = \{X \in \mathfrak{L}(G) : X \subseteq U\}$, $D_2(V) = \{X \in \mathfrak{L}(G) : X \cap V \neq \emptyset\}$, where U and V run over all open subsets of G . The structure of topological groups G for which the space $\mathfrak{L}(G)$ or some of its subspaces satisfy various compactness conditions was studied in [1-7]. At the same time, the topological structure of the space $\mathfrak{L}(G)$ itself or its subspaces, in particular, compact ones, has essentially never been studied. For instance, a characterization of compact spaces homeomorphic to $\mathfrak{L}(G)$ for a suitable topological group G is unknown (A. V. Arkhangel'skii's problem).

The present note the results of which have been announced in [8] obtains a compactness criterion for subspaces of $\mathfrak{L}(G)$ for an arbitrary locally compact group G . The structure of compact subspaces in the subspaces $\mathfrak{K}(G)$ and $n\mathfrak{K}(G)$ of, respectively, compact and noncompact subgroups of $\mathfrak{L}(G)$ is studied, along with the question when is the topology of the space $\mathfrak{L}(G)$ determined by the family of its compact sets. All studied groups are assumed to be locally compact, and the word "subgroup" refers only to closed subgroups.

LEMMA 1. If $\mathfrak{F}$ is a closed countably compact subspace in $\mathfrak{K}(G)$, then the subspace $K = \bigcup_{F \in \mathfrak{F}} F$ is compact in G .

Proof. We will first show that $\overline{K}$ is compact. Suppose the contrary. Then there exist a compact neighborhood U of the identity of the group G and a sequence $\{x_n\}$ of elements of K such that $x_i U \cap x_j U = \emptyset$ for $i \neq j$. For each natural number n we choose a subgroup $F_n \in \mathfrak{F}$ in such a way that $x_n \in F_n$. By the countable compactness of $\mathfrak{F}$, the sequence of subgroups $\{F_n\}$ has a limit point P . Since $\mathfrak{F}$ is closed $P \in \mathfrak{F}$, therefore, P is a compact subgroup. By construction, the subspace X consisting of the elements of the sequence $\{x_n\}$ is discrete and closed in G . Thus, $X \cap PU$ is finite. So there exists a natural number m such that $x_n \notin PU$ for all $n > m$. But then the neighborhood $D_1(PU)$ of the subgroup P does not contain any subgroups F_n , $n > m$. This contradicts the fact that P is a limit point of the sequence $\{F_n\}$. It remains to show that K is closed. Suppose that $x \in \overline{K}$ and a directed set $\{\alpha\}$, $\alpha \in J$, of elements of K converges to x . For each $\alpha \in J$ we choose a subgroup $F_\alpha \in \mathfrak{F}$ in such a way that $x_\alpha \in F_\alpha$. Since $\overline{K}$ is compact, it follows from Vietoris's theorem (p. 52 in [9]) that one can choose in the directed set $\{F_\alpha\}$, $\alpha \in J$, a directed subfamily $\{F_\beta\}$, $\beta \in I$, converging to some subgroup S . Since $\mathfrak{F}$ is closed, $S \in \mathfrak{F}$ and, therefore, $S \subseteq K$. It follows from the definition of the E -topology and the construction of the family $\{F_\beta\}$, $\beta \in I$, that $x \in S$. Thus, $x \in K$ and the lemma is proved.

LEMMA 2. Let $\mathfrak{F} \subseteq \mathfrak{L}(G)$, $H \in \mathfrak{F} \cap n\mathfrak{K}(G)$. If H has a countably compact neighborhood in $\mathfrak{F}$ or the character of H in $\mathfrak{F}$ is countable, then $D_1(H) \cap \mathfrak{F}$ is a neighborhood of the subgroup H in $\mathfrak{F}$.

Proof. Let $\mathfrak{U}$ be a countably compact neighborhood of the subgroup H in $\mathfrak{F}$. Suppose that the conclusion of the lemma is false. Then $H \in \mathfrak{F}_1$, where $\mathfrak{F}_1 = \mathfrak{U} \setminus D_1(H)$. By Lemma 2 in [3], $\mathfrak{F}_1$ contains an infinite discrete subspace closed in $\mathfrak{L}(G)$, contrary to the countable compactness of $\mathfrak{F}$.

Suppose the character of H in $\mathfrak{F}$ is countable and $\{\mathfrak{U}_n\}$ is a base of neighborhoods of H in $\mathfrak{F}$. Suppose that the conclusion of the lemma is false and choose in each neighborhood $\mathfrak{U}_n$ a subgroup A_n such that $A_n \not\subseteq H$. Then the sequence $\{A_n\}$ converges to H . Therefore, the subspace $\mathfrak{F}_2$ whose elements are the subgroups A_n does not contain infinite discrete subsets closed in $\mathfrak{L}(G)$. Since $H \in \overline{\mathfrak{F}_2}$, an application of Lemma 2 in [3] produces a contradiction.

THEOREM 1. A subspace $\mathfrak{F}$ in $\mathfrak{L}(G)$ is compact if and only if the following conditions are fulfilled:

- 1) $\mathfrak{F}$ is closed;
- 2) $\mathfrak{F}$ contains no infinite descending chains of noncompact subgroups;
- 3) each closed subspace $\mathfrak{F}'$ of $\mathfrak{F}$ contains only finitely many maximal in $\mathfrak{F}'$ noncompact subgroups;
- 4) if a closed subspace $\mathfrak{F}'$ of $\mathfrak{F}$ contains no noncompact subgroups, then $\bigcup_{F \in \mathfrak{F}'} F$ is compact in G .

Proof. Necessity. Let $\mathfrak{F}$ be a compact subspace in $\mathfrak{L}(G)$. Condition 1 is obvious. Suppose that condition 2 is violated and $\mathfrak{F}$ contains an infinite chain $F_1 \supset F_2 \supset \dots$ of noncompact subgroups. Since $\mathfrak{F}$ is compact, the sequence $\{F_n\}$ has a limit point F . It follows from the definition of the E-topology that $F \subseteq F_n$ for each n , and this contradicts Lemma 2. Suppose that condition 3 is not fulfilled and the subset $\mathfrak{F}'$ of noncompact subgroups maximal in $\mathfrak{F}'$ is infinite. Since $\mathfrak{F}'$ is compact, the subset $\mathfrak{F}'$ has a limit point $H \in \mathfrak{F}'$. By Lemma 2, there exists a neighborhood $\mathfrak{U}$ of the subgroup H in $\mathfrak{F}'$, consisting of subgroups contained in H . Choose two distinct subgroups F_1 and F_2 in $\mathfrak{U}$. Then $F_1 \subseteq H$, $F_2 \subseteq H$. Since $H \in \mathfrak{F}'$, the maximality of the subgroups F_1 and F_2 in $\mathfrak{F}'$ implies that $F_1 = F_2$. This contradicts the choice of the subgroups F_1 and F_2 . Finally, condition 4 follows from Lemma 1.

Sufficiency. If $\mathfrak{F} \subseteq \mathfrak{L}(G)$, then the compactness of $\mathfrak{F}$ follows immediately from conditions 1, 4, and Vietoris' theorem. Suppose that $\mathfrak{F}$ contains noncompact subgroups and put $\mathfrak{F}_0 = \mathfrak{F}$. Condition 1 and Lemma 3 in [1] imply that each noncompact subgroup in $\mathfrak{F}_0$ is contained in a maximal subgroup in $\mathfrak{F}_0$. By condition 3, the number of such subgroups is finite. Let $F_1, \dots, F_n$ be the noncompact subgroups maximal in $\mathfrak{F}_0$. Consider an arbitrary covering of $\mathfrak{F}_0$ with open sets and choose among them $\mathfrak{U}_1, \dots, \mathfrak{U}_n$ such that $F_1 \in \mathfrak{U}_1, \dots, F_n \in \mathfrak{U}_n$. Put $\mathfrak{F}_1 = \mathfrak{F}_0 \setminus (\mathfrak{U}_1 \cup \dots \cup \mathfrak{U}_n)$. If $\mathfrak{F}_1 \subseteq \mathfrak{L}(G)$, then, by the above, $\mathfrak{F}_1$ is compact and, thus, $\mathfrak{F}_0$ is also compact. Otherwise, by a similar argument, we construct $\mathfrak{F}_2$, etc. We will show that for some natural number m $\mathfrak{F}_m \subseteq \mathfrak{L}(G)$ and, therefore, $\mathfrak{F}_0$ is compact. Suppose the contrary. Then the set $\mathfrak{N}_m$ of maximal, in $\mathfrak{F}_m$, noncompact subgroups is nonempty and, by condition 3, finite for each natural number m . Note that for each subgroup K_m in $\mathfrak{N}_m$ there exists a chain of subgroups $K_m \subset K_{m-1} \subset \dots \subset K_0$, where $K_i \in \mathfrak{N}_i$, $i = 0, 1, \dots, m$. So some subgroup $N_0 \in \mathfrak{N}_0$ is the end of infinitely many such chains. Among subgroups of $\mathfrak{N}_1$ contained in N_0 there exists a subgroup N_1 through which infinitely many chains pass. Continuing this argument, we construct an infinite descending chain of subgroups $N_0 \supset N_1 \supset \dots$, contrary to condition 2. The theorem is proved.

COROLLARY 1. A subspace $\mathfrak{F}$ in $\mathfrak{L}(G)$ is compact if and only if $\mathfrak{F}$ is closed and countably compact.

Proof. The necessity is obvious. We will prove the sufficiency using Theorem 1. Condition 1 is obvious. Condition 2 follows from Lemma 2 and condition 4 follows from Lemma 1. Suppose that condition 3 is violated and $\mathfrak{F}'$ is the infinite subset of maximal in $\mathfrak{F}'$ noncompact subgroups. By virtue of its countable compactness, the subset $\mathfrak{F}'$ has a limit point H . Since $\mathfrak{F}'$ is closed, $H \in \mathfrak{F}'$. By Lemma 2, $\mathfrak{F}' \cap D_1(H)$ is a neighborhood of the subgroup H in $\mathfrak{F}'$. Thus, there exist subgroups $F_1, F_2 \in \mathfrak{F}'$, such that $F_1 \neq F_2$, $F_1 \subseteq H$, $F_2 \subseteq H$. Since the subgroups F_1 and F_2 are maximal in $\mathfrak{F}'$, it follows that $F_1 = F_2$, i.e., we have reached a contradiction.

COROLLARY 2. If $\mathfrak{F}$ is a compact subspace of $n\mathfrak{L}(G)$, then $|\mathfrak{F}| \leq \omega(G)$, where $\omega(G)$ is the weight of the group G .

Proof. Suppose the contrary and put $\mathfrak{F}_1 = \mathfrak{F}$. Since $\mathfrak{F}_1$ is compact, there exists a subgroup $F_1 \in \mathfrak{F}_1$, each neighborhood of which has cardinality greater than $\omega(G)$. By Lemma 2, $\mathfrak{F}_1 \cap D_1(F_1)$ is a neighborhood of the subgroup F_1 in $\mathfrak{F}_1$. Thus, $|\mathfrak{F}_1 \cap D_1(F_1)| > \omega(G)$. Let $\{U_\alpha, \alpha \in I\}$ be a base of topology of the subgroup F_1 and $|I| \leq \omega(G)$. Put $\mathfrak{U}_\alpha = \{H \in \mathfrak{F}_1 \cap D_1(F_1) : H \cup U_\alpha = \emptyset\}$. Clearly, $\mathfrak{F}_1 \cap D_1(F_1) = \left(\bigcup_{\alpha \in I} \mathfrak{U}_\alpha \right) \cup \{F_1\}$. Since $|\mathfrak{F}_1 \cap D_1(F_1)| > \omega(G)$, there exists $\beta \in I$, such that $|\mathfrak{U}_\beta| >$

$\omega(G)$. Since $\mathfrak{U}_\beta$ is closed in $\mathfrak{F}$, $\mathfrak{U}_\beta$ is compact. Put $\mathfrak{F}_2 = \mathfrak{U}_\beta$ and choose a subgroup F_2 each neighborhood of which in $\mathfrak{F}_2$ has cardinality greater than $\omega(G)$. Continuing this argument, we construct a descending chain of subgroups $F_1 \supset F_2 \supset \dots$, which contradicts Theorem 1.

A topological space is said to be sparse (in Cantor's sense) if each nonempty subspace of it contains a point isolated in this subspace.

COROLLARY 3. Each compact set $\mathfrak{F}$ in $n\mathfrak{R}(G)$ is sparse.

Proof. By Theorem 1, $\mathfrak{F}$ contains no infinite descending chains of subgroups. So each nonempty subspace $\mathfrak{F}_1$ of $\mathfrak{F}$ contains a subgroup F_1 minimal in $\mathfrak{F}_1$. By Lemma 2, F_1 is an isolated point of the subspace $\mathfrak{F}_1$.

Sparse compact spaces possess many remarkable properties although the problem of their classification has not been solved (cf. [10]). In connection with Corollary 3, the following question arises: is it possible to embed each sparse compact space homeomorphically in the subspace $n\mathfrak{R}(G)$ for an appropriate group G ? Spaces of ordinals admit such a realization. Moreover, each compact space in the Vietoris exponent $\exp X$ of a discrete space X can be embedded in $n\mathfrak{R}(G)$. It is unknown, however, whether each sparse compact space can be realized in $\exp X$.

It is worth noting that, for a suitable choice of a group G , each given compact space can be realized in the space $\mathfrak{L}(G)$. This follows from Tikhonov's theorem and the fact that the space $\mathfrak{L}(G)$ of the group $G = T \times C_2$ contains a segment (T is the one-dimensional torus, C_2 is the cyclic group of order 2, $\times$ is the symbol of the topological semidirect product).

The results above demonstrate a substantial difference in topological properties between the subspaces $\mathfrak{R}(G)$ and $n\mathfrak{R}(G)$. We will give some other examples. $\mathfrak{R}(G)$ is locally compact for each group G (p. 54 in [9]); $\mathfrak{R}(G)$ is σ -compact if the group G is σ -compact [2]; $\mathfrak{R}(G)$ satisfies the first countability axiom if the group G does. None of these statements is, generally speaking, true for the subspace $n\mathfrak{R}(G)$. It follows from Lemma 2 in [2] that $n\mathfrak{R}(G)$ satisfies the first countability axiom if G is a group of countable weight. Thus, for group G of countable weight both subspaces $\mathfrak{R}(G)$ and $n\mathfrak{R}(G)$ satisfy the first countability axiom. However, as the following theorem shows, this statement is not, generally, true for the space $\mathfrak{L}(G)$. A special case of Theorem 2 is the key Lemma 3 of [6].

THEOREM 2. If the space $\mathfrak{L}(G)$ of a zero-dimensional group G is locally countably compact or satisfies the first countability axiom, then each noncompact inductively compact subgroup of G is open.

Proof. Let H be an inductively compact subgroup in $n\mathfrak{R}(G)$. By Lemma 2, $D_1(H)$ is a neighborhood of H in $\mathfrak{L}(G)$. By virtue of its inductive compactness, H belongs to the closure in the E -topology of the sets of its compact subgroups. Thus, $D_1(H)$ is a neighborhood of some compact subgroup K . By definition of the E -topology, there exist subsets $U, V_1, \dots, V_n$ open in G such that $K \subseteq U, K \cap V_1 \neq \emptyset, \dots, K \cap V_n \neq \emptyset$ and the neighborhood $\mathfrak{U} = D_1(U) \cap D_2(V_1) \cap \dots \cap D_2(V_n)$ of the subgroup K is contained in $D_1(H)$.

Suppose that W is an open compact subgroup in G and $KW \subseteq U$. We choose a subgroup W_1 in such a way that $x^{-1}W_1x \subseteq W$ for each $x \in K$. We denote by W_2 the least subgroup containing $\bigcup_{x \in K} x^{-1}W_1x$. Then $P = KW_2$ is an open subgroup and $P \in \mathfrak{U}$. Thus, $F \subseteq H$ and H is open. The theorem is proved.

A natural class of topological spaces containing locally compact spaces as well as spaces with the first countability axiom is given by the k -spaces, i.e., spaces whose topology is determined by a family of compact sets (p. 153 in [11]). It follows from the above that for a group G of countable weight $\mathfrak{R}(G)$ and $n\mathfrak{R}(G)$ are k -spaces. Is $\mathfrak{L}(G)$ a k -space in this case? For connected solvable Lie groups it is so [7]. An affirmative answer for a class of groups containing all discrete and inductively pronilpotent groups of countable weight is given by Theorem 3 whose statement and proof are preceded by two lemmas.

LEMMA 3. If G is a group of countable weight, then the character of a point H in $\mathfrak{L}(G)$ in the subspace $D_1(H)$ is countable.

The proof follows from the definition of a base of the E -topology.

LEMMA 4. Suppose that $K \subseteq H$ is a subgroup of a group G and K is compact and open in H . If $\bigcup_{\alpha \in I} x_\alpha K$ is the decomposition of H into cosets relative to K , then there exists a neighborhood U of the identity of the group G such that $x_\alpha KU \cap x_\beta KU = \emptyset$ for $\alpha \neq \beta$.

Proof. Let V be an open subset in G with $V \cap H = K$. Using the compactness of K , we choose neighborhoods W and U of the identity of the group G such that $KW \subseteq V$ and $xUU^{-1}x^{-1} \subseteq W$.

for each $x \in K$. Suppose that $x_\alpha KU \cap x_\beta KU \neq \emptyset$. Then $x_\alpha k_1 u_1 = x_\beta k_2 u_2$ for some $k_1, k_2 \in K$, $u_1, u_2 \in U$. Thus, $x_\beta^{-1} x_\alpha = k_2 u_2 u_1^{-1} k_1^{-1} = k_2 k_1^{-1} (k_1 u_2 u_1^{-1} k_1^{-1}) \in k_2 k_1^{-1} W \subseteq KW \subseteq V$. On the other hand, $x_\beta^{-1} x_\alpha \in H$. Since $V \cap H = K$, we have $x_\beta^{-1} x_\alpha \in K$, i.e., $\alpha = \beta$.

THEOREM 3. If G is a group of countable weight and the set of its inductively compact subgroups is closed in $\mathfrak{L}(G)$, then $\mathfrak{L}(G)$ is a k -space.

Proof. Suppose the contrary and choose a subset $\mathfrak{F}$ not closed in $\mathfrak{L}(G)$ such that the intersection of $\mathfrak{F}$ with each compact set in $\mathfrak{L}(G)$ is closed. Take an arbitrary subgroup $F \in \overline{\mathfrak{F}} \setminus \mathfrak{F}$. Since $\mathfrak{K}(G)$ is open in $\mathfrak{L}(G)$ and locally compact, $F \in n\mathfrak{K}(G)$. We consider two cases.

- 1) $F \in \overline{\mathfrak{F} \cap n\mathfrak{K}(G)}$. By Lemma 2 in [2], $F \in \overline{(\mathfrak{F} \cap n\mathfrak{K}(G)) \cap D_1(F)}$. Using Lemma 3, we choose a sequence $\{F_n\}$ of subgroups in $\mathfrak{F} \cap n\mathfrak{K}(G) \cap D_1(F)$ converging to F . The subspace $\mathfrak{F}'$ whose elements are F and the members of the sequence $\{F_n\}$ is compact and $F \in \overline{\mathfrak{F} \cap \mathfrak{F}'}$. Thus, $F \in \mathfrak{F}$, contrary to its choice.
- 2) $F \in \overline{\mathfrak{F} \cap \mathfrak{K}(G)}$. In order to obtain a contradiction, it suffices to show, as before, that $F \in \overline{(\mathfrak{F} \cap \mathfrak{K}(G)) \cap D_1(F)}$. Suppose that it is not true and choose, using the regularity of $\mathfrak{L}(G)$, a neighborhood $\mathfrak{U}$ of the subgroup F such that $\overline{\mathfrak{U} \cap (\mathfrak{F} \cap \mathfrak{K}(G)) \cap D_1(F)} = \emptyset$. Since $F \in \mathfrak{K}(G)$, F is inductively compact by the hypothesis of the theorem. We write F in the form $\bigcup_{n=1}^{\infty} F_n$, where F_n are compact subgroups open in F and $F_n \subset F_{n+1}$ for each n .

We partition F into cosets relative to $F_1: F = \bigcup_{i=1}^{\infty} x_i F_1$. Using Lemma 4, choose a neighborhood U of the identity of the group G such that $x_i F_1 U \cap x_j F_1 U = \emptyset$ for $i \neq j$.

We will construct, inductively, a sequence $\{U_n\}$ of compact neighborhoods of the identity of the group G in such a way that $D_1(F_1 U_1 \cup \dots \cup F_n U_n) \cap (\mathfrak{F} \cap \overline{\mathfrak{U}}) = \emptyset$ and $U_n \subseteq U$ for each n . First, we will define U_1 . Suppose that for each neighborhood V of the identity of the group G $D_1(F_1 V) \cap \mathfrak{F} \cap \overline{\mathfrak{U}} \neq \emptyset$. It follows from Vietoris' theorem that $D_1(F_1) \cap \overline{\mathfrak{F} \cap \mathfrak{U}} \neq \emptyset$. Note that $D_1(F_1) \subseteq D_1(F) \cap \mathfrak{K}(G)$ and $\overline{\mathfrak{F} \cap \mathfrak{K}(G)} \subseteq \mathfrak{F}$. Thus, $\emptyset \neq D_1(F_1) \cap \overline{\mathfrak{F} \cap \mathfrak{U}} \subseteq \overline{\mathfrak{U} \cap (D_1(F) \cap \mathfrak{K}(G) \cap \mathfrak{F})}$, which contradicts the choice of the neighborhood $\mathfrak{U}$ and guarantees the possibility of defining U_1 .

Suppose that the neighborhoods $U_1, \dots, U_n$ have been constructed but the neighborhood U_{n+1} cannot be chosen, i.e., $D_1(F_1 U_1 \cup \dots \cup F_n U_n \cup F_{n+1} U') \cap (\mathfrak{F} \cap \overline{\mathfrak{U}}) \neq \emptyset$ for each neighborhood U' of the identity of G . Then, by Vietoris' theorem, there exists a subgroup $S \in D_1(F_1 U_1 \cup \dots \cup F_n U_n \cup F_{n+1}) \cap (\mathfrak{F} \cap \overline{\mathfrak{U}})$. Since $S \in \overline{\mathfrak{F} \cap \mathfrak{K}(G)} \subseteq \mathfrak{F} \cap \mathfrak{K}(G)$, we have $S \not\subseteq F_1 U_1 \cup \dots \cup F_n U_n$ by the induction hypothesis. Let $x \in (F_{n+1} \setminus F_n) \cap S$ and let y be an arbitrary element in $(F_1 U_1 \cup \dots \cup F_n U_n) \cap S$. Then $y = fu$, where $f \in F_k$, $u \in U_k$, $k \leq n$. Furthermore, $x^{-1}y = (x^{-1}f)u \in (F_{n+1} \setminus F_n) U_k \subseteq (F_{n+1} \setminus F_n) U$. On the other hand, $x^{-1}y \in S \subseteq F_n U \cup (F_{n+1} \setminus F_n)$. By the choice of the neighborhood U , $(F_{n+1} \setminus F_n) U \cap F_n U = \emptyset$. Thus, $x^{-1}y \in F_{n+1} \setminus F_n$ and $y \in F_{n+1}$, i.e., $S \subseteq F_n$. Thus, we have proved that $S \in (D_1(F_{n+1}) \cap \mathfrak{F} \cap \mathfrak{K}(G)) \cap \overline{\mathfrak{U}}$, contrary to the choice of the neighborhood $\mathfrak{U}$. This completes the inductive construction.

Consider the neighborhood $\mathfrak{W} = D_1\left(\bigcup_{i=1}^{\infty} F_i U_i\right) \cap \overline{\mathfrak{U}}$ of the subgroup F . Since $F \in \overline{\mathfrak{F} \cap \mathfrak{K}(G)}$, there exists a subgroup $K \in \mathfrak{F} \cap \mathfrak{K}(G)$ such that $K \in \mathfrak{W}$. In particular, $K \subset \bigcup_{i=1}^{\infty} F_i U_i$. Since K is compact there exists a natural number m such that $K \subseteq \bigcup_{i=1}^m F_i U_i$. This contradicts the fact that $D_1 \times (F_1 U_1 \cup \dots \cup F_m U_m) \cap \mathfrak{F} \cap \overline{\mathfrak{U}} = \emptyset$. The theorem is proved.

COROLLARY 4. The space $\mathfrak{L}(G)$ of a group G of countable weight is a k -space if one of these conditions is fulfilled:

- 1) G contains an open compact normal subgroup;
- 2) the inductively compact radical of the group G contains all compact elements of G .

Proof. In the former case the condition that the set of inductively compact subgroups is closed in $\mathfrak{L}(G)$ follows from Lemma 1 in [2], in the latter case it is obvious.

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SMOOTHNESS IN THE PARAMETER OF AN INVARIANT TORUS OF A QUASILINEAR SYSTEM OF DIFFERENTIAL EQUATIONS

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Many problems of the theory of nonlinear oscillations reduce to a system of differential equations of the form

$$d\psi/dt = \omega + \varepsilon\Delta + \varepsilon f_1(h) + \varepsilon^2 F_1(\psi, h, \Delta, \varepsilon), \quad dh/dt = \varepsilon Hh + \varepsilon f_2(h) + \varepsilon^2 F_2(\psi, h, \Delta, \varepsilon), \quad (1)$$

whose right side is defined in the domain

$$\psi \in \mathcal{J}_m, \quad \|h\| \leq \delta, \quad \|\Delta\| \leq \sigma_0, \quad \varepsilon \in [0, \varepsilon_0] \quad (2)$$

for sufficiently small positive δ , ε_0 , is periodic in $\psi = (\psi_1, \dots, \psi_m)$ with period 2π , ε times continuously differentiable in ψ , $h = (h_1, \dots, h_n)$, $\Delta = (\Delta_1, \dots, \Delta_{n_0})$, ε and satisfies the conditions

$$f_1(0) = 0, \quad f_2(0) = 0, \quad \partial f_2(0)/\partial h = 0. \quad (3)$$

Here $\mathcal{J}_m$ is the cube of periods of a function periodic in ψ with period 2π .

An invariant surface of the system (1) of the form

$$h = u(\psi, \Delta, \varepsilon), \quad \psi \in \mathcal{J}_m, \quad (4)$$

periodic in ψ_j , $j = \overline{1, m}$, with period 2π is called an invariant torus of the system of equations (1) of smoothness equal to the number of derivatives with respect to ψ of the function $u(\psi, \Delta, \varepsilon)$. The theory of perturbations of invariant tori of systems of the form (1) (cf., e.g., [1-4]) guarantees the existence of an invariant torus (4) under the assumption that the real parts of the eigenvalues of the matrix H , assumed to be constant, are different from zero. It follows from these results that for finite ε one can always fix ε_0 so that the